

Summation

COP 3502C – Computer Science I

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Summation /1

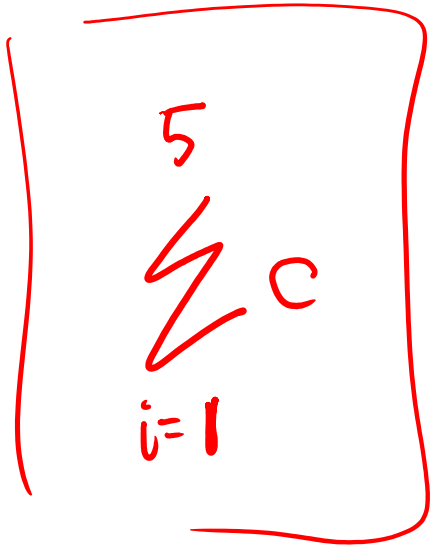
- Process of adding things together
- Essentially, that is what we have been doing
- A compact notation where the capital Greek letter Sigma is used

$$\sum_{i=1}^n f(i)$$

Summation /2

$$C + C + C + C + C$$

If you are repeatedly adding a constant value **c**



```
3  ☐ int get_sum(int start, int end) {  
4      ☐ int total = 0;  
5      ☐ int c = 1;  
6  
7      ☐ for(int i = start; i <= end; i++) {  
8          total = total + c;  
9      }  
10  
11     return total;  
12 }
```

⚡ Summation /3

$$1 + 2 + 3 + 4 + \dots + n$$

If you are adding a series of numbers

$$\begin{matrix} n \\ \swarrow \searrow \\ i \\ i=1 \end{matrix}$$

```
3  [ ] int get_sum(int start, int end) {  
4  [ ]   int total = 0;  
5  [ ]  
6  [ ]   for(int i = start; i <= end; i++) {  
7  [ ]       total = total + i;  
8  [ ]   }  
9  [ ]  
10 [ ]   return total;  
11 [ ] }
```

Summation /4

- Because our goal is to perform an informal Big-Oh analysis, we have to find a way to end up with a mathematical expression
- In short, we are looking for a **closed-form** of this summation
- **Note:** No need to memorize because a formula sheet will be provided. However, you need to manipulate your summation for you to be able to correctly use them.

Closed Form Summations

- Basic Summation

- $\sum_{k=m}^n c = c + c + \dots + c = (n - m + 1)c = O(n)$

- Arithmetic Series

- $\sum_{k=1}^n k = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} = O(n^2)$

- Sum of Squares

- $\sum_{k=1}^n k^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6} = O(n^3)$

- Sum of Cubes

- $\sum_{k=1}^n k^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4} = O(n^4)$

- Geometric Series

- $\sum_{k=0}^n x^k = 1 + x + x^2 + x^3 + \dots + x^n = \frac{x^{n+1}-1}{x-1} = \frac{1-x^{n+1}}{1-x} = O(x^n)$

- If the summation is infinite and not a constant or variable

- $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$

Summation Formulas

$$\sum_{i=1}^n c = cn \quad \sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}, \quad \sum_{i=0}^{\infty} x^i = \frac{1}{1-x}, \quad |x| < 1$$

Useful Summations Manipulations

- Our goal is to simplify our summations to find the closed form.
- The following formulas are useful in manipulating summations that can generally lead to a simple summation and its closed form.
- $\sum c a_k = c \sum a_k$
- $\sum (a_k + b_k) = \sum a_k + \sum b_k$

$$\sum_{i=1}^n \underline{5x}$$

$$\underbrace{5x + 5x + 5x}_{n}$$

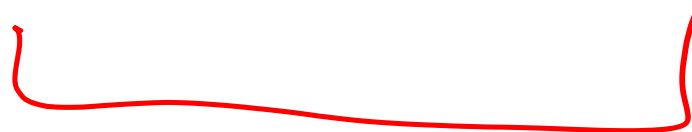
$$= 5 \times n$$

Summation Formulas

$$\sum_{i=1}^n c = cn, \sum_{i=1}^n i = \frac{n(n+1)}{2}, \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}, \sum_{i=0}^{\infty} x^i = \frac{1}{1-x}, |x| < 1$$

$$\sum_{i=1}^{3n} 5$$

$$5 + 5 + 5 + 5$$



$$3n$$

$$= 5 \cdot 3n$$

$$= 15n$$

Summation Formulas

$$\sum_{i=1}^n c = cn \quad \sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}, \quad \sum_{i=0}^{\infty} x^i = \frac{1}{1-x}, \quad |x| < 1$$

$$\sum_{i=1}^{3n} i$$

$$(1 + 3n) \left[\frac{3n}{2} \right]$$

$$\frac{(1 + 3n) 3n}{2}$$

$$\frac{3n + 9n^2}{2}$$

Summation Formulas

$$\sum_{i=1}^n c = cn, \sum_{i=1}^n i = \frac{n(n+1)}{2}, \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}, \sum_{i=0}^{\infty} x^i = \frac{1}{1-x}, |x| < 1$$

$$\sum_{i=1}^{3n} n$$

$$\underbrace{n + n + n + \dots}_{3n}$$

$$n \cdot 3n = \boxed{3n^2}$$

Summation Formulas

$$\sum_{i=1}^n c = cn \quad \sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}, \quad \sum_{i=0}^{\infty} x^i = \frac{1}{1-x}, \quad |x| < 1$$

$$\sum_{i=1}^{2n} (2i - 2n)$$

$$= \sum_{i=1}^{2n} 2i - \sum_{i=1}^{2n} 2n$$

$$\frac{2(1+2+3+4)}{2+4+6+8}$$

$$2+4+6+8$$

$$2 \sum_{i=1}^{2n} i - [(2n)(2n)]$$

$$- 4n^2$$

$$2 \left[\frac{2n(2n+1)}{2} \right] - 4n^2$$

$$2n(2n+1) - 4n^2 = 4n^2 + 2n - 4n^2$$

$$\boxed{2n}$$

Summation Formulas

$$\sum_{i=1}^n c = cn \quad \sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}, \quad \sum_{i=0}^{\infty} x^i = \frac{1}{1-x}, \quad |x| < 1$$

$$\sum_{i=1}^n \sum_{j=1}^m c$$

$$\sum_{j=1}^m c$$

$$c + c + c + \dots + c$$

m

$$\sum_{i=1}^n mc$$

$$mc + mc + \dots + mc$$

n

$$= n \cdot m \cdot c$$

Summation Formulas

$$\sum_{i=1}^n c = cn, \sum_{i=1}^n i = \frac{n(n+1)}{2}, \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}, \sum_{i=0}^{\infty} x^i = \frac{1}{1-x}, |x| < 1$$

$$\sum_{i=1}^n \left[\sum_{j=1}^i c \right]$$

$$\sum_{j=1}^i c \Rightarrow ci$$

$$\sum_{i=1}^n ci \Rightarrow c \sum_{i=1}^n i$$

$$c \left[\frac{(n+1)n}{2} \right]$$

Summation Formulas

$$\sum_{i=1}^n c = cn, \sum_{i=1}^n i = \frac{n(n+1)}{2}, \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}, \sum_{i=0}^{\infty} x^i = \frac{1}{1-x}, |x| < 1$$

$$\sum_{i=1}^n \sum_{j=1}^i (2i)$$

$$\sum_{j=1}^i 2i$$

$$= 2i \cdot i$$

$$= 2i^2$$

$$\sum_{i=1}^n 2i^2 \Rightarrow 2 \sum_{i=1}^n i^2 \Rightarrow \frac{1}{2} \left[\frac{n(n+1)(2n+1)}{3} \right]$$

Summation Formulas

$$\sum_{i=1}^n c = cn \quad \sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}, \quad \sum_{i=0}^{\infty} x^i = \frac{1}{1-x}, \quad |x| < 1$$

$$\frac{n(n+1)(2n+1)}{3}$$

$$\sum_{i=k}^n i$$

5

$$5+6+7+8+9+10$$

=

$$[1+2+3+4+5+6+7+8+9+10]$$

$$- [1+2+3+4]$$

$$[5+6+7+8+9+10]$$

$$\sum_{i=k}^n i = \sum_{i=1}^n i - \sum_{i=1}^{k-1} i$$

$$\frac{n(n+1)}{2} - \left[\frac{(k-1)(k-1+1)}{2} \right]$$

$$\left[\frac{n(n+1)}{2} - \frac{(k-1)k}{2} \right]$$

Summation Formulas

$$\sum_{i=1}^n c = cn, \sum_{i=1}^n i = \frac{n(n+1)}{2}, \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}, \sum_{i=0}^{\infty} x^i = \frac{1}{1-x}, |x| < 1$$

$$\sum_{i=10}^{80} (i^3 + i^2)$$

Summation Formulas

$$\sum_{i=1}^n c = cn \quad \sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}, \quad \sum_{i=0}^{\infty} x^i = \frac{1}{1-x}, \quad |x| < 1$$

$$\sum_{i=10}^{80} (i^3 + i^2) = \underbrace{\sum_{i=10}^{80} i^3}_{\text{red}} + \underbrace{\sum_{i=10}^{80} i^2}_{\text{red}}$$

$$\left[\sum_{i=1}^{80} i^3 - \sum_{i=1}^9 i^3 \right] + \left[\sum_{i=1}^{80} i^2 - \sum_{i=1}^9 i^2 \right]$$

$$\left[\frac{80^2(80+1)^2}{4} - \frac{9^2(9+1)^2}{4} \right] + \left[\frac{80(80+1)(2(80)+1)}{6} - \frac{9(9+1)(2(9)+1)}{6} \right]$$

I don't have access to a calculator
so this is my answer.

Summation Formulas

$$\sum_{i=1}^n c = cn \quad \sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}, \quad \sum_{i=0}^{\infty} x^i = \frac{1}{1-x}, |x| < 1$$

$$\sum_{i=10}^n (i^3 + i^2 - 2)$$

Summation Formulas

$$\sum_{i=1}^n c = cn \quad \sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}, \quad \sum_{i=0}^{\infty} x^i = \frac{1}{1-x}, \quad |x| < 1$$

$$\sum_{i=10}^n (i^3 + i^2 - 2) \Rightarrow \sum_{i=10}^n i^3 + \sum_{i=10}^n i^2 - \sum_{i=10}^n 2$$

$$\left[\sum_{i=1}^n i^3 - \sum_{i=1}^9 i^3 \right] + \left[\sum_{i=1}^n i^2 - \sum_{i=1}^9 i^2 \right] - \left[\sum_{i=1}^n 2 - \sum_{i=1}^9 2 \right]$$

$$\left[\frac{n^2(n+1)^2}{4} - \frac{9^2(9+1)^2}{4} \right] + \left[\frac{n(n+1)(2n+1)}{6} - \frac{9(9+1)(2(9)+1)}{6} \right] - [2n - 18]$$

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Summation Formulas

$$\sum_{i=1}^n c = cn, \sum_{i=1}^n i = \frac{n(n+1)}{2}, \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}, \sum_{i=0}^{\infty} x^i = \frac{1}{1-x}, |x| < 1$$

Final Notes

- Known closed-form formulas for summations usually assume that the index starts at 1
- If a summation begins at a different value, we must first rewrite or shift the summation so that it starts at 1 before applying the formula.

Questions?